

IT24103868

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IT2120- Probability and Statistics

Lab – 10

```
setwd("C:\\Users\\Asus\\Desktop\\IT24103868")
```

```
##Q1
```

```
observed <- c(120, 95, 85, 100)
prob <- c(.25, .25, .25, .25)
chisq.test(x=observed, p=prob)
```

```
> setwd("C:\\Users\\Asus\\Desktop\\IT24103868")
> ##Q1
>
> observed <- c(120, 95, 85, 100)
> prob <- c(.25, .25, .25, .25)
> chisq.test(x=observed, p=prob)
```

Chi-squared test for given probabilities

```
data: observed
X-squared = 6.5, df = 3, p-value = 0.08966
```

```
##Q2
```

```
file_path <- "http://www.sthda.com/sthda/RDoc/data/housetasks.txt"
housetasks <- read.delim(file_path, row.names = 1)
housetasks
chisq <- chisq.test(housetasks)
chisq
```

```

> file_path <- "http://www.sthda.com/sthda/RDoc/data/housetasks.txt"
>
> housetasks <- read.delim(file_path, row.names = 1)
> housetasks

```

	Wife	Alternating	Husband	Jointly
Laundry	156	14	2	4
Main_meal	124	20	5	4
Dinner	77	11	7	13
Breakfeast	82	36	15	7
Tidying	53	11	1	57
Dishes	32	24	4	53
Shopping	33	23	9	55
Official	12	46	23	15
Driving	10	51	75	3
Finances	13	13	21	66
Insurance	8	1	53	77
Repairs	0	3	160	2
Holidays	0	1	6	153

```

>
> chisq <- chisq.test(housetasks)
> chisq

```

Pearson's Chi-squared test

```

data:  housetasks
X-squared = 1944.5, df = 36, p-value < 2.2e-16

```

##Q3

```

#Consider 25% level of significant for the class|
#Rejection Region: if the p value for the test is less than 0.25,
#Reject the null hypothesis at 25% level of significant
#P value for the test got as 2.2e-16

```